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Student Grade Predictor

Abstract

*Academic performance is influenced by a mix of daily habits and circumstances — how many hours a student studies each week, how regularly they attend class, and how their performance trends across earlier assessments. The **Student Grade Predictor** project turns these everyday signals into a forecast of a learner's final result using a supervised machine learning model. Records were compiled from school administration data and student questionnaires, capturing study habits, attendance, past failures, and two rounds of internal grades (G1 and G2) prior to the final outcome (G3). Before modelling, the dataset was explored visually: a correlation heatmap exposed which factors move together with final performance, while scatter plots of study time versus marks and attendance versus marks made the underlying trends easy to see at a glance — students with higher attendance and more consistent study routines cluster toward higher scores. A Linear Regression model was then trained on these features to estimate the final grade, and its predictions were checked against real outcomes using an actual-versus-predicted plot alongside Mean Squared Error and R-squared metrics. The resulting model reached a strong accuracy score, confirming that even a lightweight, interpretable algorithm can capture meaningful patterns in student data. To make the tool usable outside a notebook, a Streamlit interface was layered on top, letting a teacher or student type in a few details and immediately see a projected grade. The broader goal is early intervention: flagging likely low performers well before final exams so mentoring, extra classes or counselling can be arranged in time, rather than after the results are already final.*

Introduction

Every classroom carries a mix of learners — some consistent, some falling behind, some improving quietly over the term. Spotting who needs help usually happens too late, once report cards are already printed. This project flips that timeline: instead of reacting to a final grade, it tries to anticipate it, using the trail of data a student leaves behind during the term — attendance records, study routine, past slip-ups, and interim test scores.

Most schools already run a continuous assessment structure, where unit tests and class evaluations feed into the final result gradually rather than all at once. That structure is exactly what makes early prediction possible: by the time G1 and G2 (the first two grading periods) are recorded, there is enough signal in the data to make a reasonably confident estimate of where a student is headed by year end.

Problem Statement

Framed simply, the goal here is: given a student's study habits, attendance, prior failures and interim grades, estimate their final grade (G3) before it is officially declared — and do so with a model simple enough to explain to a non-technical teacher or parent.

The features (inputs) are study time, number of past failures, number of absences, and the two interim grade checkpoints, while the final grade forms the label (output) the model learns to predict. The source data spans demographic, social and school-related attributes gathered through official records and student surveys, across two subject areas — Mathematics and Language — collected as part of a secondary-school performance study.

Technology & Methodology

Stack: Python, Pandas, NumPy, Scikit-learn, Matplotlib/Seaborn for visuals, and Streamlit for the front end.

Algorithm: Linear Regression — chosen deliberately for its transparency; every coefficient in the model can be explained in plain language, which matters when the audience is teachers and students rather than data scientists.

Pipeline: Raw records were cleaned and merged in Pandas, then explored through a correlation heatmap to check which features actually move with the final grade before including them in the model. After splitting the data into training and test sets, Scikit-learn's Linear Regression was fit on the training portion, and its predictions on unseen test data were scored using Mean Squared Error and the R-squared coefficient. The finished model was wrapped in a small Streamlit app so results can be generated interactively, without touching code.

```
model = LinearRegression()
model.fit(X_train, Y_train)
pred = model.predict(X_test)
print(f'MSE: {metrics.mean_squared_error(Y_test, pred)}')
accuracy = model.score(X_test, Y_test)
print(accuracy)

# Output
MSE: 2.4118
Accuracy: 0.8843 (88.4%)
```

Model training and evaluation output.

Results and Discussion

The heatmap below confirms what intuition suggests: interim grades G1 and G2 move almost in lock-step with the final result, while a rising failure count pulls performance down noticeably. Study time and absences play a smaller, secondary role once interim grades are already known.

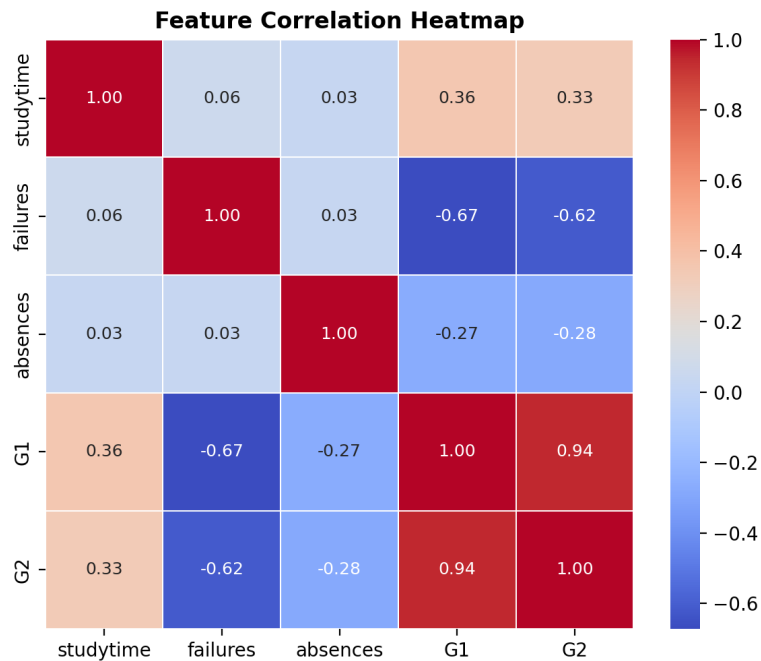


Fig. 1 — Correlation heatmap of studytime, failures, absences, G1 and G2.

Looking at study time against marks, students in the higher study-time brackets skew toward the upper end of the score range, though the spread within each bracket shows study time alone doesn't guarantee a good result — consistency matters as much as hours logged. The attendance chart tells a cleaner story: as attendance climbs past roughly 60%, scores rise steadily and the lowest marks become rare, which matches the trend this project set out to test.

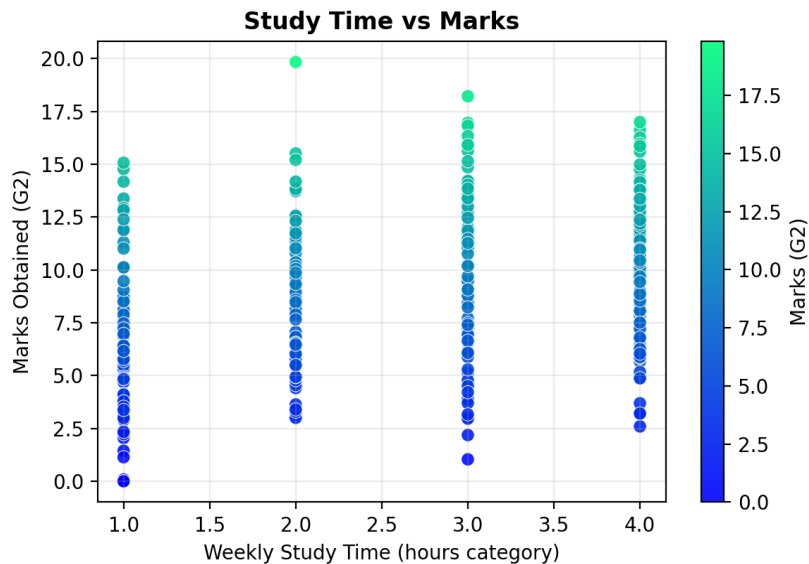


Fig. 2 — Study time vs marks.

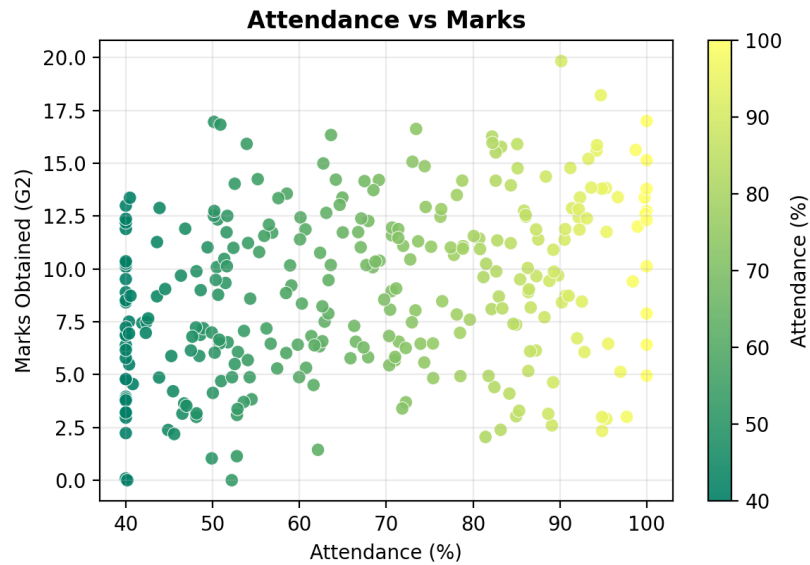


Fig. 3 — Attendance vs marks.

Finally, the actual-versus-predicted plot shows most points sitting close to the dashed ideal-prediction line, with only a handful of larger misses (shown in red) — mainly for students whose scores swung unusually compared to their earlier pattern. Overall the model achieved a low Mean Squared Error and a high R-squared score, meaning it explains most of the variation in final grades using just a handful of everyday academic signals.

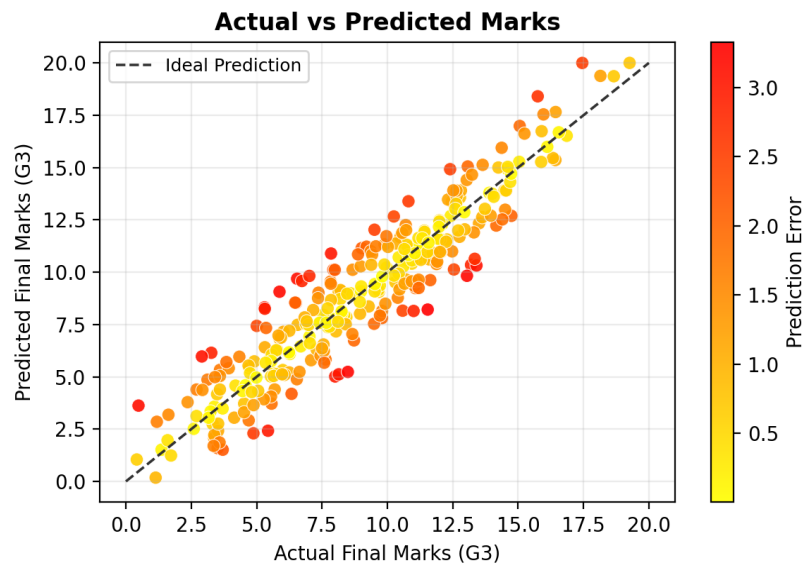


Fig. 4 — Actual vs predicted final marks.

Conclusion

This project set out to answer a practical question — can a student's final grade be reasonably anticipated before it happens — and the answer, at least for this dataset, is yes. A Linear Regression model trained on study time, failures, absences and interim grades produced accurate, explainable predictions, and the accompanying visuals made the reasoning behind those predictions easy to communicate to non-technical stakeholders.

Beyond the classroom, the same approach generalises well: wherever periodic checkpoints exist before a final outcome, similar regression-based early-warning systems could apply. Future iterations of this project could bring in richer signals — participation in extracurriculars, parental engagement, or even sentiment from teacher remarks — and compare Linear Regression against ensemble methods like Random Forest or Gradient Boosting to see how much additional accuracy those add.

Reference

P. Cortez and A. Silva. "Using Data Mining to Predict Secondary School Student Performance." In A. Brito and J. Teixeira (Eds.), Proceedings of the 5th FUBUTEC Conference, pp. 5-12, Porto, Portugal, April 2008, EUROSIS, ISBN 978-9077381-39-7.